LECTURE-4

UML Diagram

Application → ① Paragraphs
→ ② Diagram

UML

Structural
(Static)

There are 7 structural diagram

↳ Class Diagram
(99% Interview imp. question)

Behavioural
(Dynamic)

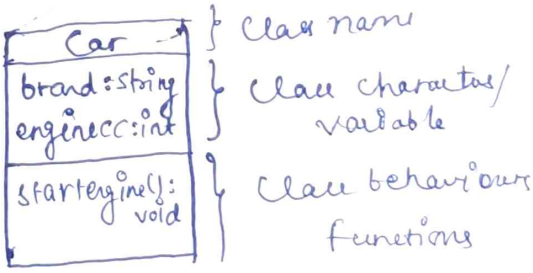
There are 7 behavioural diagram.

↳ Sequence Diagram

⇒ **Class Diagram –**

(1) Class structure
(2) Association / Connection

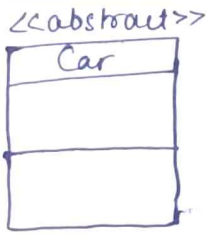
Q. How to make class?



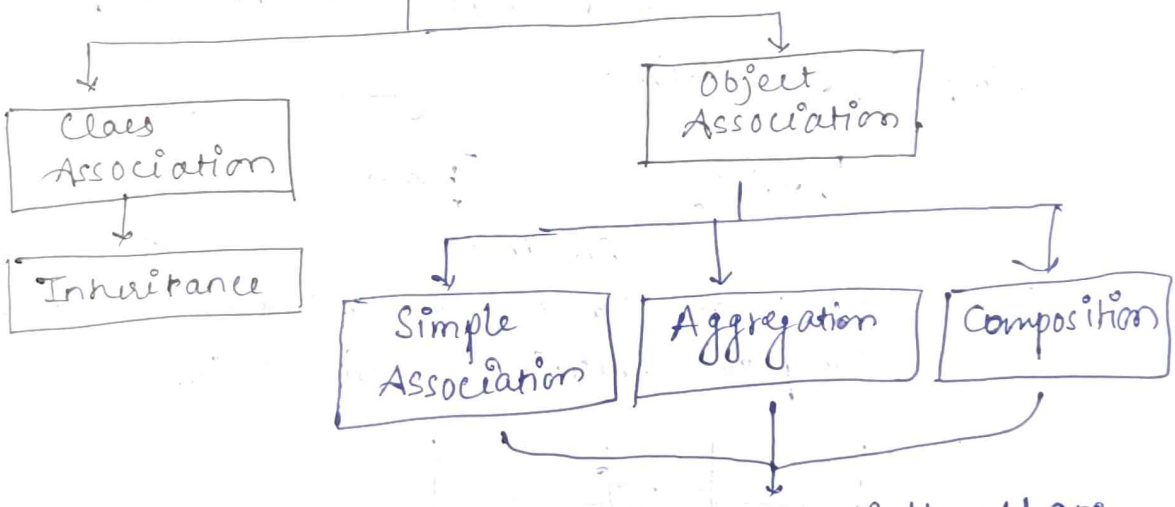
Access Modifiers

	Within class	From child class	Outside class	Symbol in Diagram
Public	(✓)	(✓)	(✓)	'+'
Protected	(✓)	(✓)	(×)	'#'
Private	(✓)	(×)	(×)	'–'

⇒ if class is 'abstract class' then-



Associations

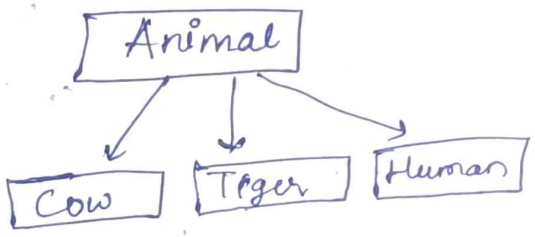


Programmatically all are Composition but in theory their relations are different

Inheritance → (is-a) relationship

(⟶ is-a)

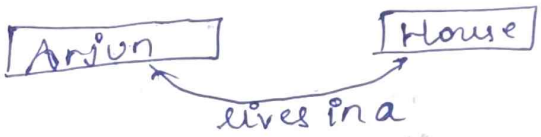
```
class A {
 method1();
   }

class B:public A {
  method2();
 }
main() {
   B *b = new B();
     b→method1();
     b→method2();
   }
```



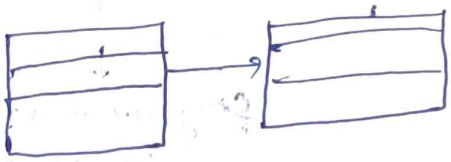
Cow is a Animal

Composition - → (has-a) relationship

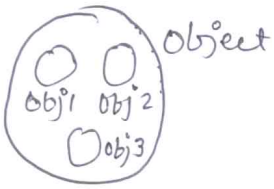
Simple Association - → [⟶] representation

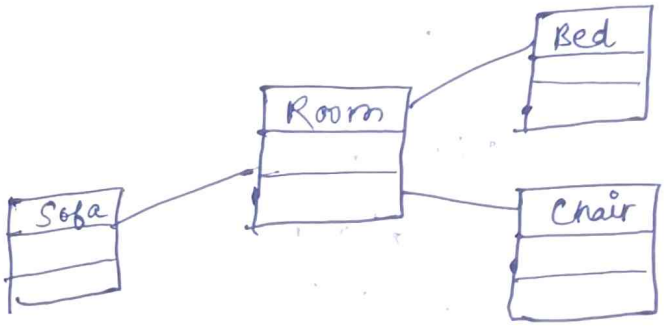


Two objects or classes are related but not a complex relation there is just a link.

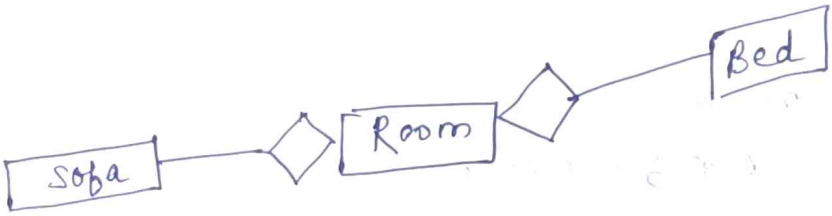


Aggregation – Some more complex then simple association.





Symbol/Representation

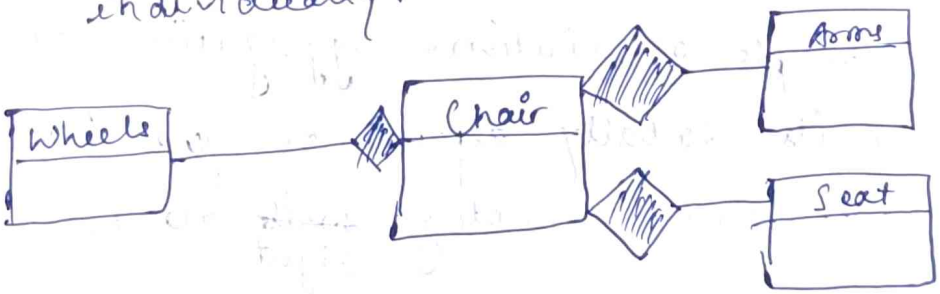


Composition – Strongest form of relationship.

↳ handles complex relationships

Symbol/Representation

In this every dependent entity can't exist individually.



Here wheels, Arms, seat can't exists individually.

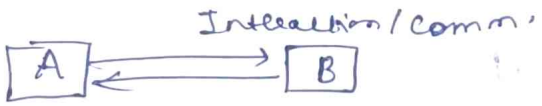
```
Class A {
    method1();
}

class B {
    A *a;
  B() {
    a = new A();}

    method2();
}

    main() {
      B *b = new B()
      b→ method2();
      b→ a→ method1();
    }
```

} to call method1 from b without inheritance is composition here.

→ Sometimes there are thin lines between simple association, aggregation and composition so its totally depends on you how you take it there is nothing ~~wrong~~ right and wrong.

Sequence Diagram

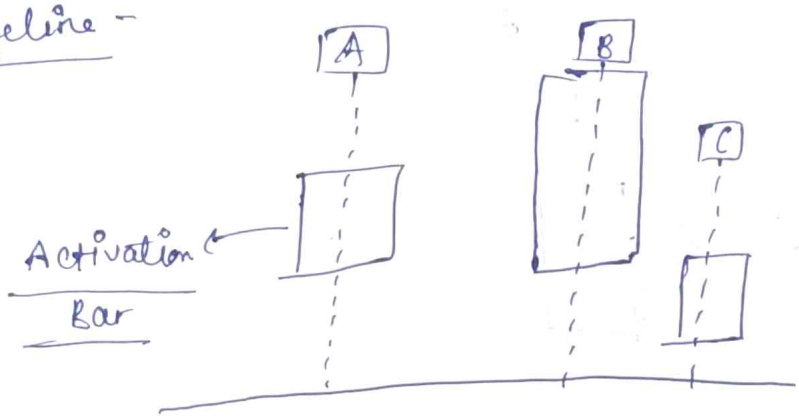
It tells about communication between two entities.



→ Representation of class -

} normally in box

→ Lifeline -

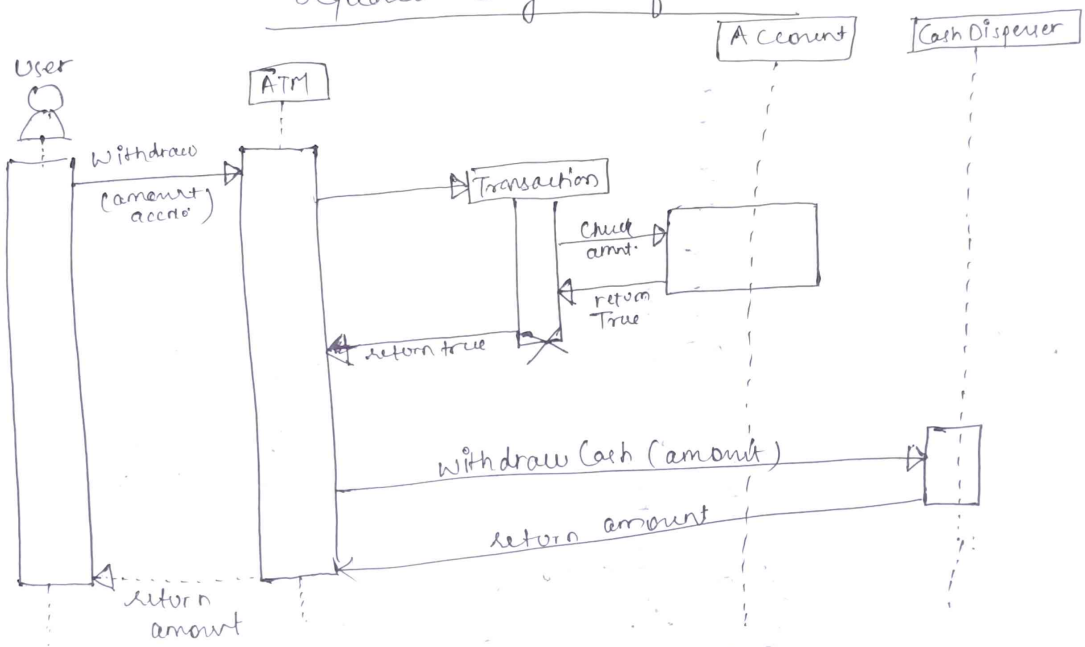


Here we say that A and B class starts with application and exists whenever application exists while class C comes in middle.

→ Activation Bar - tells active period time of a class.

→ Messages -
- → Asynchronous Message
- → Synchronous Message

Sequence Diagram of ATM



alt : denotes (if-else)

Option : (if)

loop : for / while